and, since $\bar{\boldsymbol{Q}}$ is known, this gives

$$\gamma = \left(\frac{\mathrm{E} - \bar{U}}{\bar{K}} \right)^{1/2}, \tag{3.13}$$

where $\bar{U} = U(\bar{Q})$ and $\bar{K} = \bar{\boldsymbol{P}}^T \boldsymbol{M}^{-1} \bar{\boldsymbol{P}}/2$ are the potential and kinetic energies after a step of the original non-conserving scheme. As long as the steps are small enough we could hope that the resulting method is sensible. This is, however, not quite true. If, in the harmonic oscillator, $\bar{p}_{n+1}$ happens to vanish, then γ is not defined. We could work around this obstacle by assuming a large number of degrees of freedom, in which case $\bar{K}$ is only zero if all the momenta simultaneously vanish and this situation is, in a realistic model of a molecule, extremely unlikely. We also must assume that $\mathrm{E} - \bar{U} \geq 0$, which, for a large system, seems the likely situation, since we may suppose $\bar{K} + \bar{U} = \bar{\mathrm{E}} \approx \mathrm{E}$, thus $\mathrm{E} - \bar{U} \approx \bar{K} \geq 0$. Thus we assume a system with many degrees of freedom so that all the conditions for the method to be well defined are satisfied. Although it may appear that the method requires two evaluations of the energy at each timestep (which may seem a high cost, since Euler's method and even the 2nd order Verlet method only require one), the evaluation of the energy needed to compute γ is at the end point of the combined step, and this is the starting point for the next step—essentially the same work would be needed anyway to compute the forces at the next step. In practice, some care must be exercised in implementing this method. In particular the kinetic energy needs to be monitored for approach to zero; when this occurs the projections must be disabled or replaced by an alternate scheme.

When we adjust the momenta by scaling in order to satisfy the energy constraint, we are, in effect, projecting the preliminary numerical approximation onto an energy surface, using one particular type of projection. There are many alternative projection methods which we could use for this purpose, which might alter both the positions and momenta. Modifying the positions means that we will somehow need to solve the equation

$$H(\boldsymbol{Q}, \boldsymbol{P}) = \mathrm{E},$$

where $\boldsymbol{Q}$ depends on a parameter or parameters (typically a Lagrange multiplier that is used to maintain the constraint). These methods are necessarily implicit and require solving nonlinear equations involving the potential energy U [308]. In an iterative strategy such as Newton's method, we will need to compute the potential energy several times at each timestep. This is a very high price to pay in practice and is one reason why complicated energy projection methods are not popular.

Another reason why energy conserving methods are rarely used in molecular simulation is that they may lead to behavior that is not consistent with the properties of our original model. If we think of the procedure that we use to define the step as defining a map of phase space, the map does not necessarily have qualitative properties that mimic those of the original system. Note that, as we have mentioned